

Weibel Introduction to Homological Algebra Solutions

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This is a solution set to the first few problems in Weibel's "Introduction to Homological Algebra". I plan to type of some more of them soon. Solutions are brief and generally aim to give the main ideas.

Chain Complexes

1.1.1

It is a quick verification that composing two maps gives the 0 map ($4 \cdot 4 = 16 \equiv 0 \pmod{8}$). Now for $n \geq 1$ the kernel of each map is $2\mathbb{Z}$ and the image into each C_n is $4\mathbb{Z}$ so the homology $H_n(C)$ is $\mathbb{Z}/2$. For $n = 0$ the kernel of each map is $\mathbb{Z}$ and the image into it is $\mathbb{Z}/4$ so the homology is $\mathbb{Z}/4$. For everything else the kernel and homology are trivial.

1.1.2

We only need to consider a small section of the diagram.

$$\begin{array}{ccccc} C_{n+1} & \longrightarrow & C_n & \longrightarrow & C_{n-1} \\ \downarrow u_{n+1} & & \downarrow u_n & & \downarrow u_{n-1} \\ D_{n+1} & \longrightarrow & D_n & \longrightarrow & D_{n-1} \end{array}$$

We claim that u_n sends boundaries to boundaries and cycles to cycles. Note that the image of a boundary is a boundary since $u_n \circ d = d \circ u_{n+1}$. Furthermore the image of a cycle is a cycle since $u_{n-1} \circ d = d \circ u_n$.

1.1.3

Note that the composition of two maps is 0 since we have $C_n \rightarrow B_{n-1} \subset C_{n-1} \rightarrow B_{n-2}$ however B_{n-1} and B_{n-2} do not intersect in C_{n-1} . Now suppose we have a chain sequence of vector spaces

$$\cdots \rightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots$$

Then write:

$$0 \rightarrow \ker(d_n) \rightarrow C_n \rightarrow \operatorname{im}(d_n) \rightarrow 0 \quad 0 \rightarrow \operatorname{im}(d_{n+1}) \rightarrow \ker(d_n) \rightarrow H_n(C) \rightarrow 0$$

Note that vector spaces are projective, therefore they split in exact sequences. We can therefore write $C_n = \ker(d_n) \oplus \operatorname{im}(d_n)$ and $\ker(d_n) = \operatorname{im}(d_{n+1}) \oplus H_n(C)$. Putting this together we obtain $C_n = \operatorname{im}(d_n) \oplus \operatorname{im}(d_{n+1}) \oplus H_n(C)$.

1.1.4

Consider the chain complex

$$\cdots \rightarrow \text{Hom}_R(A, C_{n+1}) \rightarrow \text{Hom}_R(A, C_n) \rightarrow \text{Hom}_R(A, C_{n-1}) \rightarrow \cdots$$

Where the boundary maps are $\phi \mapsto d \circ \phi$. Then the composition of two boundary maps has $\phi \mapsto d \circ d \circ \phi = 0 \circ \phi = 0$ as desired.

Now take $A = Z_n$. Suppose that $H_n(C) \neq 0$. Then $Z_n \neq B_n$. Consider the inclusion map $\phi : Z_n \subseteq C_n$ which is an element of $\text{Hom}_R(A, C_n)$. This map is a boundary of $\text{Hom}_R(A, C_n)$ since it is 0 when postcomposed with d . Furthermore it cannot be the image of a map in $\psi \in \text{Hom}_R(A, C_{n+1})$ since the image of such a $d \circ \psi$ is strictly contained in B_n .

Now consider the converse assertion. We have an exact sequence

$$0 \rightarrow C_{n+1} \rightarrow C_n \rightarrow C_{n-1} \rightarrow 0$$

which will not generally be exact after we apply $\text{Hom}_R(A, C)$ since this functor is not left exact. For instance take

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

and note that $Z_n = \mathbb{Z}$. Now note that

$$\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z} \quad \text{Hom}(\mathbb{Z}, \mathbb{Q}) \cong \mathbb{Q} \quad \text{Hom}(\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) = 0$$

Therefore we obtain the sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow 0$$

which is nonsensical since $\mathbb{Z}$ is not isomorphic to $\mathbb{Q}$.

1.1.5

Note that (1) and (2) are equivalent by the definition of exactness. Furthermore the condition of the map $0 \rightarrow C$ being a quasi-isomorphism is by definition equivalent to all the $H_n(C)$ being 0.

1.1.6

We will first address $H_0(C)$. Note the the kernel of the map $C_0 \rightarrow C_{-1} = 0$ is C_0 . The map $d : C_1 \rightarrow C_0$ map a linear combination of edges $e_{ij} : v_i \rightarrow v_j$ to $ke_{ij} \mapsto k(v_i - v_j)$. This will exactly induce the relation $v_i = v_j$ in the quotient space. Therefore if the graph is connected all the edges will be "equal" to eachother in the quotient and the space will have dimension 1.

Now we will address $H_1(C)$. The image of $0 = C_2 \rightarrow C_1$ is 0 so we are asked to find the dimension of the kernel of the map $C_1 \rightarrow C_0$. Note that the matrix has V rows and E columns, and has an image of dimension 1. Therefore by rank nullity theorem the dimension of the kernel is $V - E - 1$. This corresponds to the number of circuits since each circuit essentially corresponds to a linearly independent element of the null space. IE if we have a circuit

$$v_1 \rightarrow v_2 \rightarrow \cdots \rightarrow v_n \rightarrow v_1$$

then the element $e_{12} + e_{23} + \cdots + e_{n-1,n} + e_{n1}$ is an element of the null space.

1.1.7

We can label the vertices $0, 1, 2, 3$ (I will not draw a picture since the choice is irrelevant). Each face is given by removing one of the vertices, so we call the orientation of the i -th face (which is removing the i -th vertex) $(-1)^i$ with respect to the induced orientation of the larger simplex. More generally, given an edge of a simplex, its orientation is defined with respect to which vertices we removed and the order we removed them. To determine the orientation of a general "sub-simplex", first write all the omitted vertices *in the order they were omitted*, then write all the vertices it contains according to the ambient order of the larger simplex. The sign of this permutation is the orientation of the edge. This show that $d^2 = 0$ since for $n > 1$ we have that S_n has an equal number of odd and even permutations.

Now we write the maps. I will denote $s_{a_1, \dots, a_n}$ as the sub simplex spanned by the $a_1, \dots, a_n$ vertices. Then a direct computation gives the matrices:

	s_{012}	s_{013}	s_{023}	s_{123}		s_{01}	s_{02}	s_{03}	s_{12}	s_{13}	s_{23}
s_{01}	1	1	0	0		s_0	-1	-1	-1	0	0
s_{02}	-1	0	1	0		s_1	1	0	0	-1	-1
s_{03}	0	-1	-1	0		s_2	0	1	0	1	0
s_{12}	1	0	0	1		s_3	0	0	1	0	1
s_{13}	0	1	0	-1							
s_{23}	0	0	1	1							

1.2.1

We claim that in an abelian category maps $A \rightarrow C$ and $B \rightarrow D$ induce maps of the biproducts $A \times B \rightarrow C \times D$. Furthermore this preserves exactness. This is because a map out of the biproduct is specified by maps out of each summand. Now compose by the natural injections

$$A \rightarrow C \rightarrow_{l_C} C \times D \quad B \rightarrow D \rightarrow_{l_D} C \times D$$

which specifies a map $A \times B \rightarrow C \times D$ by the other universal property of the biproduct.

Now consider exact sequences of the form

$$0 \rightarrow A \rightarrow C \rightarrow E \rightarrow 0 \quad 0 \rightarrow B \rightarrow D \rightarrow F \rightarrow 0$$

and construct the sequence:

$$0 \rightarrow A \times B \rightarrow C \times D \rightarrow E \times F \rightarrow 0$$

We may test for exactness by applying the Hom functors.

$$\text{Hom}(A \times B, P) \rightarrow \text{Hom}(C \times D, P) \rightarrow \text{Hom}(E \times F, P) \rightarrow 0$$

$$\text{Hom}(P, E \times F) \rightarrow \text{Hom}(P, C \times D) \rightarrow \text{Hom}(P, A \times B) \rightarrow 0$$

and all these Homs decompose as direct product by the universal property. We have essentially reduced to problem to the right exactness of biproduct on abelian groups. This is by a diagram chase.

$$G_1 \times G_2 \rightarrow G_3 \times G_4 \rightarrow G_5 \times G_6 \rightarrow 0$$

every element of the kernel is by given by some $g_3 \times g_4$ with $g_3 \mapsto 0$ and $g_4 \mapsto 0$. This gives g_1 and g_2 so that $g_1 \times g_2 \mapsto g_3 \times g_4$ by exactness of the two sequences.

Then we

$$0 \rightarrow \text{im}(d_{n-1}^\alpha) \rightarrow \ker(d_n^\alpha) \rightarrow H_n(A_\alpha) \rightarrow 0$$

$$0 \rightarrow \text{im}(d_{n-1}^\beta) \rightarrow \ker(d_n^\beta) \rightarrow H_n(A_\beta) \rightarrow 0$$

Firstly note that At this point we note that homology is an abelian category, so we can take the biproduct of the exact sequences.

$$0 \rightarrow \text{im}(d_{n-1}^\alpha) \times \text{im}(d_{n-1}^\beta) \rightarrow \ker(d_n^\alpha) \times \ker(d_n^\beta) \rightarrow H_n(A_\alpha) \times H_n(A_\beta) \rightarrow 0$$

1.2.2

We will consider the map $h : A \rightarrow B$.

First of all, every kernel $K \rightarrow_k A \rightarrow_h B$ is mono. This is by the universal property. Pick $f, g : C \rightarrow K$. Then both f and g correspond uniquely maps $\tilde{f} : C \rightarrow A$ and $\tilde{g} : C \rightarrow A$. These correspond to $k \circ f$ and $k \circ g$. Therefore k is left cancellative.

Similarly cokernels $A \rightarrow_h B \rightarrow_c C$ are epis. Choose $f, g : C \rightarrow D$. Then both f and g correspond uniquely to maps $B \rightarrow D$ which corresponds to $f \circ c$ and $g \circ c$. Therefore this map is right cancellative and is an epi.

Now suppose that $f : A \rightarrow B$ and using the abelian condition we construct the kernel of its cokernel

$$\begin{array}{ccccc} & \text{ker coker } f & & & \\ & \downarrow u & \searrow h & & \\ A & \xrightarrow{f} & B & \xrightarrow{g} & \text{coker}(f) \end{array}$$

Note that $g \circ f = 0$ and therefore since $\text{ker coker } f$ is universal with respect to this property we get a map $u : \text{ker coker } f \rightarrow A$ with $u \circ f = h$. We that u is always an isomorphism when f is monic. Similarly if we construct the maps

$$\begin{array}{ccccc} & & & \text{coker ker } f & \\ & & \nearrow h & \downarrow u & \\ \text{ker } f & \xrightarrow{g} & A & \xrightarrow{f} & B \end{array}$$

We see again that u must be an isomorphism when f is epi.

1.2.3

Consider the diagram of maps:

$$\begin{array}{ccccccc} & \text{ker}(f_{n+1}) & & \text{ker}(f_n) & & \text{ker}(f_{n-1}) & \\ & \downarrow \iota_{n+1} & & \downarrow \iota_n & & \downarrow \iota_{n-1} & \\ \cdots & \longrightarrow & A_{n+1} & \xrightarrow{d_{n+1}} & A_n & \xrightarrow{d_n} & A_{n-1} \longrightarrow \cdots \\ & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} & \\ \cdots & \longrightarrow & B_{n+1} & \xrightarrow{d_{n+1}} & B_n & \xrightarrow{d_n} & B_{n-1} \longrightarrow \cdots \\ & \downarrow \pi_{n+1} & & \downarrow \pi_n & & \downarrow \pi_{n-1} & \\ & \text{coker}(f_{n+1}) & & \text{coker}(f_n) & & \text{coker}(f_{n-1}) & \end{array}$$

We need to define the maps on the top and the bottom. Consider the composition $d_{n+1} \circ \iota_{n+1} : \text{ker}(f_{n+1}) \rightarrow A_n$. By commutativity of the diagram this map is killed by precomposition by f_n . Therefore by the universal property of the kernel it gives a map $d_{n+1} : \text{ker}(f_{n+1}) \rightarrow \text{ker}(f_n)$. The composition of two of these maps $d_n \circ d_{n+1}$ must be zero by left cancellativity after postcomposing with ι_{n-1} (which gives the 0 map by commutativity).

Now we will define the maps on the bottom. The map $d_{n+1} : B_{n+1} \rightarrow B_n$ is killed by precomposition by f_{n+1} and therefore gives a map $\text{coker}(f_{n+1}) \rightarrow B_n$ which extends to a map $\text{coker}(f_{n+1}) \rightarrow \text{coker}(f_n)$ after postcomposing with π_n . The composition of two of these maps $d_n \circ d_{n+1}$ must be zero by right cancellativity after precomposition with π_{n+1} (which gives 0 by commutativity).

1.2.4

The statement that

$$0 \rightarrow A \rightarrow_a B \rightarrow_b C \rightarrow 0$$

Is exactly the statement that

- (a) a is monic.
- (b) b is an epi.
- (c) $\ker a = \operatorname{coker} b$.

However we showed in 1.2.3 that all these conditions are checked piecewise. Therefore this is equivalent to

$$0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$$

being exact for all n .

1.2.5

Suppose that the columns are exact (ie $\ker d_k^v = \operatorname{im} d_{k+1}^v$) and consider a section of the diagram:

$$\begin{array}{ccccc}
 & C_{p-1,q+1} & & & \\
 & \downarrow d^v & & & \\
 & C_{p-1,q} & \xleftarrow{d^h} & C_{p,q} & \\
 & \downarrow d^v & & \downarrow d^v & \\
 C_{p-1,q-1} & \xleftarrow{d^h} & C_{p,q-1} & \xleftarrow{d^h} & C_{p+1,q-1} \\
 & & \downarrow d^v & & \downarrow 0 \\
 & & C_{p,q-2} & \xleftarrow{0} & 0
 \end{array}$$

By boundedness we may assume that $C_{p,q-1}$ has the maximum value of $p - q$ along the line $p + q = n$ (indicated by the 0). We will focus only on 3 diagonals so relabel the diagram like this:

$$\begin{array}{ccccc}
 & A_1 & & & \\
 & \downarrow d^v & & & \\
 & B_1 & \xleftarrow{d^h} & A_0 & \\
 & \downarrow d^v & & \downarrow d^v & \\
 C_1 & \xleftarrow{d^h} & B_0 & \xleftarrow{d^h} & A_{-1} \\
 & & \downarrow d^v & & \downarrow 0 \\
 & & C_0 & \xleftarrow{0} & 0
 \end{array}$$

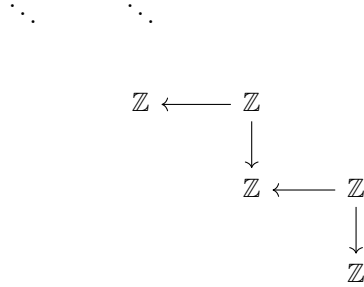
So that A, B, C are the n th, $n - 1$ th and $n - 2$ th elements of the total chain complex. Suppose that we have an element $b_0 \times b_1 \times \dots$ of the kernel. We claim that we can lift this to some $a_0 \times a_1 \times \dots$. Since there is no horizontal contribution to C_0 we have exactness at $A_0 \rightarrow B_0 \rightarrow C_0$ and we can determine a_0 so that $d(a_0)|_{B_0} = b_0$. Now we proceed via induction. Suppose we have $b_0 \times \dots \times b_{k+1}$ mapping to 0 on $C_0, \dots, C_{k+1}$. By construction $b_0 \times \dots \times b_k$ maps to 0 on $C_0, \dots, C_k$ so we lift to $a_0 \times \dots \times a_k$. Note that

$$\begin{aligned}
 d(b_0 \times \dots \times b_{k+1})|_{C_{k+1}} &= d^v(b_{k+1}) + d^h(b_k) \\
 &= d^v(b_{k+1}) + d^h(d^v(a_k) + d^h(a_{k-1})) \\
 &= d^v(b_{k+1}) + d^h(d^v(a_k)) + d^h(d^h(a_{k-1})) \\
 &= d^v(b_{k+1}) - d^v(d^h(a_k)) \\
 &= d^v(b_{k+1} - d^h(a_k))
 \end{aligned}$$

which implies that $b_{k+1} - d^h(a_k)$ lies in the kernel of d^v . Therefore we can lift it up along the exact sequence in the $k + 1$ th column to a_{k+1} . Then by construction $a_0 \times \cdots \times a_{k+1}$ is a lift of $b_0 \times \cdots \times b_{k+1}$ and this process will eventually terminate by induction.

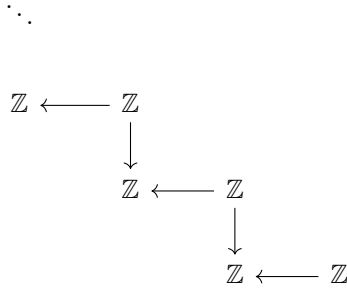
1.2.6

- (a) On the diagonal of the second quadrant make an infinite diagonal of $\mathbb{Z}$:



and fill everything else with 0. The maps between $\mathbb{Z}$ are identity. Now consider trying to map into the 1 on the bottommost $\mathbb{Z}$ and 0 everywhere else. This is only possible in the product complex with $-1, 1, -1, 1, \dots$ since one needs to cancel infinite terms.

- (b) We do a similar staircase but slightly differently:



Now, the bottommost term has a map to the left instead of an arrow into it from the top. Note that $\underbrace{1, -1, 1, -1, 1}_{k \text{ times}}$ on the upper diagonal maps to ± 1 on the k th copy of $\mathbb{Z}$ in the lower diagonal and 0 everywhere else. Therefore using this we can hit every element of the lower diagonal nonzero on finite indices. However note that the diagonal map preserves the alternating sum of the elements everywhere except at the bottommost $\mathbb{Z}$. Therefore $1, -1, -1, -1, \dots$ cannot be hit since that would require the bottommost $\mathbb{Z}$ to take infinite value.

1.2.7

For the second we see that we must show exactness of the following sequence:

Since $H_k = Z_k/B_k$ we obtain the first injection whose image is Z_k/B_k . This is by definition the kernel of the differential map. The image of the differential map is B_{k-1} inside Z_{k-1} and the cokernel map gives a surjective map to H_{k-1} which is exact.

1.2.8

$$D_n = B_{n-1} \oplus C_n \quad d = \begin{bmatrix} d_B & 0 \\ f & -d_C \end{bmatrix}$$
$$0 \rightarrow C_n \rightarrow B_{n-1} \oplus C_n \rightarrow B_{n-1} \rightarrow 0$$

1.3.1

1.3.2

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1.3.3

First we have a claim:

Claim 1.0.1 (Strong 4 Lemma). Consider the diagram

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \xrightarrow{h} & D \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \xrightarrow{h'} & D' \end{array}$$

if α is epic and δ is a monic then $g(\ker \beta) = \ker \gamma$ and $g'^{-1}(\operatorname{im} \gamma) = \operatorname{im} \beta$.

The proof is a diagram chase, I have written some diagrams below:

$$\begin{array}{ccccccc} a_7 & & b_4 & \longrightarrow & c_0 & \longrightarrow & 0_3 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ a'_6 & \longrightarrow & b'_5 & \longrightarrow & 0_1 & \longrightarrow & 0_2 \end{array} \qquad \begin{array}{ccccccc} b_4 & \longrightarrow & c_1 & \longrightarrow & 0_3 & & \\ & & \downarrow & & \downarrow & & \\ b'_0 & \longrightarrow & c'_1 & \longrightarrow & 0_2 & & \end{array}$$

The order in which the elements were constructed is marked via subscript, the reader should convince themselves that all the steps work (and understand why). In the first I start with $c \in \ker \gamma$ and note that $b - f(a)$ is the desired element in $\ker \beta$. In the second I start with $b' \in g'^{-1}(\operatorname{im} \gamma)$ and note that $\beta(b)$ differs from b' by an element $a' \in A'$ via exactness. If a is a lift of a' along α then $b - f(a)$ is the desired element of B mapping to b' .

As a corollary, if β is monic then $g(0) = \ker \gamma$ so $\ker \gamma = 0$ so γ is monic. Dually if γ is epic then then $g^{-1}(C') = \operatorname{im} \beta$ so $\operatorname{im} \beta = B'$ and β is epic. The five lemma follows from applying this result to the 2 subdiagrams.

1.3.4

The maps $H^n(C) \rightarrow H^n(B(-1))$ are trivial and factor through 0. Therefore splitting up the long exact sequence by 0s along these maps gives the result.

1.3.5

The map $f : A \rightarrow B$ splits into two short exact sequences:

$$0 \rightarrow \ker f \rightarrow A \rightarrow \operatorname{im} f \rightarrow 0 \qquad 0 \rightarrow \operatorname{im} f \rightarrow B \rightarrow \operatorname{coker} f \rightarrow 0$$

Since $\ker f$ and $\operatorname{coker} f$ are acyclic when we take the long exact sequence of cohomology we see that f is an isomorphism $H_n(A) \cong H_n(\operatorname{im} f) \cong H_n(B)$ (via f) and we are done.

1.3.6

Recall that the category of bicomplexes can be identified with the the category of chain complexes of chain complexes.